Samiksha Bhade

Node.js Backend Developer | REST APIs | WebSockets | SQL
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Summary

Node.js Backend Developer with 1.5+ years of experience building **REST APIs**, **WebSocket** services and **third-party integrations** for three live sportsbook and casino platforms, plus **JWT authentication** and loyalty modules for a live casino platform. Also built **slot game engines** in Node.js and validated them with multi-threaded **Monte Carlo RTP simulations**. Skilled in **SQL**, **MySQL**, **PostgreSQL**, **Redis**, **Docker** and **AWS**, with experience in debugging, code reviews and Agile teams.

Technical Skills

Languages: JavaScript (ES6+), SQL, Python
Backend: Node.js, Express.js, REST APIs, WebSockets (Socket.IO), JWT, Worker Threads, Async Queue Workers
Databases: MySQL, PostgreSQL, Sequelize, Drizzle ORM, Redis, Schema Design, Query Optimization
Cloud & DevOps: AWS, Docker, Git, GitHub, GitLab
Tools & Integrations: Postman, API Testing, Debugging, Code Review, Mailjet, Customer.io, SMSCountry
Game Engine & Simulation: Slot Engine Development, Par Sheet Implementation, Payline / Ways / Cluster-Pay Evaluation, Free-Spin & Buy-Feature Logic, Win Capping, RTP Simulation (Monte Carlo), Volatility Analysis, Provably Fair RNG (HMAC-SHA256)

Experience

GammaStack (iGaming Solutions) **Bengaluru, Karnataka**
Solution Engineer — Node.js Backend & Game Engine Developer *Mar 2025 – Present*

- Developed **REST APIs**, **WebSocket** systems and **queue-worker** integrations (**Mailjet, Customer.io, SMSCountry**) for **3 live sportsbook and casino platforms**.
- Optimized **SQL** across KYC, transaction and reporting modules; performed **code reviews** and resolved **production issues** in Agile teams.
- Built **slot game engines** in **Node.js** from certified specification sheets (par sheets): payline, ways-pay and cluster-pay evaluation, wild multipliers, sticky wilds, overlay-scatter retriggers, buy-feature and ante modes, and **10,000x max-win capping**.
- Fixed a **max-win cap bug** (cap applied per free spin instead of per round) and a **production crash** in free-spin bet placement caused by early loop termination on cap-hit, resolved with a defensive fallback guard.
- Converted the RTP simulation from single-threaded (∼4 hours) to a **multi-threaded harness** (**Worker Threads** across all CPU cores) running up to **1B base-game and 100M buy-feature spins**, tracking per-symbol contribution, win-distribution buckets and streaming variance to validate **RTP and volatility**.
- Converted par sheet images (reel strips, weight tables, paylines) into engine **config JSON** and used **Python (PIL)** to transcribe and diff the data, catching positional mismatches before release.
- Wrote **precision-testing** and win-range **forced-outcome (FOE)** generator scripts across base, ante and buy modes, verifying that payout rounding does not drift RTP across **BRL, EUR, USD and TRY** bet lists.

Projects

Micro Maldives Casino Platform | *Node.js, SQL, JWT, WebSockets, Customer.io, SMSCountry*

- Developed the **gamification engine, loyalty program, referral system, affiliate management** and **responsible-gambling** modules for a live casino platform.
- Implemented **JWT** authentication with email and phone verification via **SMSCountry OTP**; integrated **Customer.io** for event-driven email campaigns.
- Built real-time **WebSocket** push notifications, group chat and chat moderation.

Wild West Gold Slot Engine | *Node.js, Worker Threads, Docker, Python*

- Built a **5x4, 40-payline** engine: wilds on reels 2–4 with weighted **2x/3x/5x multipliers** summed on a payline, **sticky wilds** in free game, **overlay scatters** retriggering spins, and a **100x buy feature** on separate reel sets, capped at **10,000x**.
- Implemented base game, free game, API response and RTP simulation runner from the par sheet (**96.09% RTP**, high volatility), targeting its published hit rate and RTP.
- Fixed a hardcoded line bet that **inflated RTP ∼10x** when total bet changed, and a paytable calibrated to a fixed bet unit instead of the number of lines.

Sweet Bonanza Slot Engine | *Node.js, Worker Threads*

- Built a **cluster-pays tumble** engine with base-game and **buy-feature** modes against a certified par sheet.
- Root-caused a **∼1.4% base RTP overshoot** to scatter payout being folded into base-game payout on free-game trigger, bringing the engine back in line with the par sheet target.
- Moved the RTP simulation from a **single-threaded run of ∼4 hours** to **Worker Threads** on all CPU cores, enabling 1B base-game and 100M buy-feature spin runs.

Education

Annie Institute of Technology and Research Center **2020 – 2024**
Bachelor of Engineering in Computer Science | CGPA: 8.15/10.0 *Chhindwara, Madhya Pradesh*